Security Analysis of SHA-3 Hash Function

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Outline

Introduction

Collision Aattacks on SHA-3

Preimage Attacks on SHA-3

Attacks on Keyed Mode of Keccak

Distinguishers of Keccak

Summary

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Summary

Cryptographic hash function

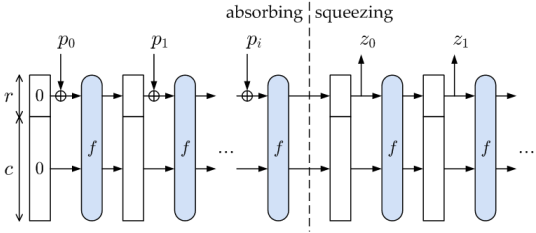
- A cryptographic hash function is a mathematical algorithm that maps data of arbitrary size to a bit string of a fixed size, which is designed to also achieve certain security properties.
- Properties
  - Collision resistance
    - It should be difficult to find a pair of different messages m_1 and m_2 such that $H(m_1) = H(m_2)$.
  - Preimage resistance
    - Given an arbitrary n-bit value x, it should be difficult to find any message m such that $H(m) = x$.
  - Second preimage resistance
    - Given message m_1, it should be difficult to find any different message m_2 such that $H(m_1) = H(m_2)$.
- Usage
  - digital signatures, message authentication codes, key derivation functions, pseudo random functions, blockchain, and many other security applications

SHA-3 hash function

- NIST SHA-3 hash function competition (2007–2012)
- Winner: Keccak
  - The winner was announced to be Keccak in October 2012.
  - Designers: Guido Bertoni, Joan Daemen, Michaël Peeters, and Gilles Van Assche
    Official versions: Keccak-224/256/384/512
    The Keccak web site: `https://keccak.team/`
- In August 2015 NIST announced that SHA-3 had become a hashing standard.
  - SHA3-224/256/384/512
  - SHAKE128/256 (eXtendable Output Functions, XOFs)

Specifications of Keccak

- Structure of Keccak
  - Sponge construction [Eurocrypt 2008, ⋆ IACR Test-of-Time Award (2023)]



- Keccak-f permutation
  - 1600 bits: a 5×5 array of 64-bit lanes
  - 24 rounds
  - each round consists of five steps:

$$\iota \circ \chi \circ \pi \circ \rho \circ \theta$$

  - χ : the only nonlinear operation

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Collision Attacks on Round-Reduced SHA-3

Methods	Keccak-224	Keccak-256	Keccak-384	Keccak-512
Differential [NRM11] SAT-based [HMR+12]	2 (practical)	2 (practical)	-	-
Differential [DDS12]	4 (practical)	4 (practical)	-	-
Internal Differential [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)

Itai Dinur, Orr Dunkelman, Adi Shamir: New Attacks on Keccak-224 and Keccak-256. FSE 2012. ⋆ Best Paper Award

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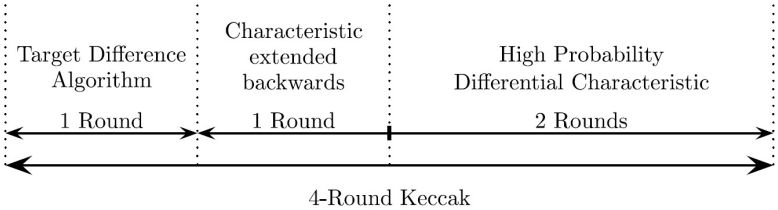
Collision Attacks on 4-round SHA-3



Itai Dinur, Orr Dunkelman, Adi Shamir: New Attacks on Keccak-224 and Keccak-256. FSE 2012. ⋆ Best Paper Award

Itai Dinur, Orr Dunkelman, Adi Shamir: Improved Practical Attacks on Round-Reduced Keccak. J. Cryptology 27(2): 183–209 (2014)

Collision Attacks on Round-Reduced SHA-3

Methods	Keccak-224	Keccak-256	Keccak-384	Keccak-512	SHAKE128
Differential or SAT-based	2 (practical)	2 (practical)	-	-	*
Differential [DDS12]	4 (practical)	4 (practical)	-	-	*
Internal Diff. [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)	-
Algebriac Diff. [QSLG17]	5 (2^{101})	-	-	-	5 (practical)

Kexin Qiao, Ling Song, Meicheng Liu, Jian Guo: New Collision Attacks on Round-Reduced Keccak. EUROCRYPT 2017.

Collision Attacks on 5-round SHA-3

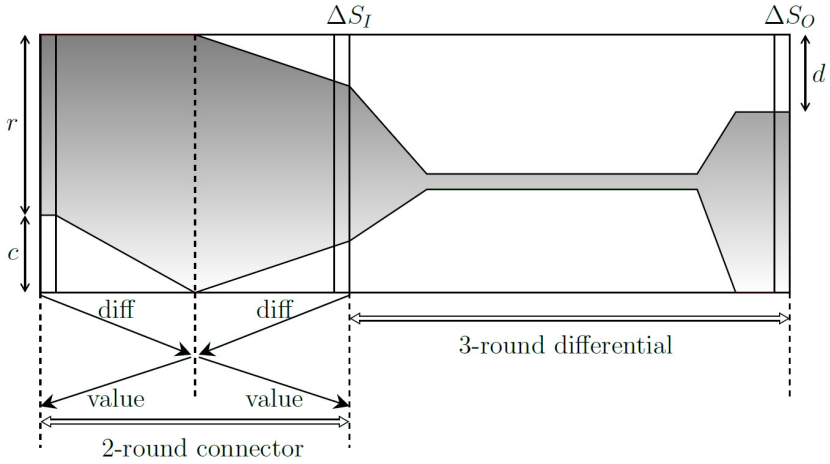


Figure: Collision attacks on 5-round Keccak

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Collision Attacks on 5-round SHA-3

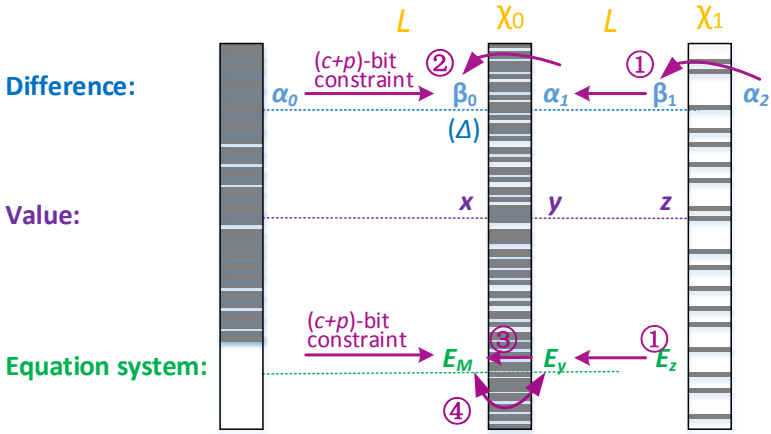

Figure: The 2-round connector in 5-round collision attacks

Kexin Qiao, Ling Song, Meicheng Liu, Jian Guo: New Collision Attacks on Round-Reduced Keccak. EUROCRYPT 2017.

Collision Attacks on Round-Reduced SHA-3

Methods	Keccak-224	Keccak-256	Keccak-384	Keccak-512	SHAKE128
Differential or SAT-based	2 (practical)	2 (practical)	-	-	*
Differential [DDS12]	4 (practical)	4 (practical)	-	-	*
Internal Diff. [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)	-
Algebriac Diff. [QSLG17]	5 (2^{101})	-	-	-	5 (practical)
Algebriac Diff. [SLG17]	5 (practical)	-	-	-	-

Ling Song, Guohong Liao, Jian Guo: Non-Full Sbox Linearization: Applications to Collision Attacks on Round-Reduced Keccak. CRYPTO 2017.

Collision attacks on 5- and 6-round Keccak

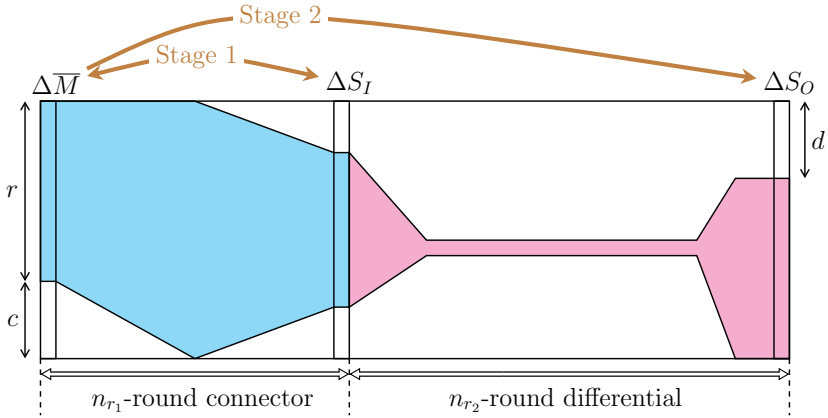

Figure: Collision attacks on 5- and 6-round Keccak

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Collision attacks on 5- and 6-round Keccak

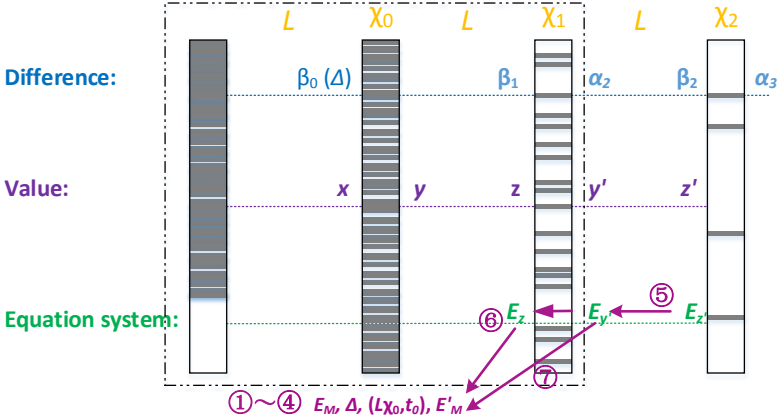

Figure: The 3-round connector in 5- and 6-round collision attacks

Ling Song, Guohong Liao, Jian Guo: Non-Full Sbox Linearization: Applications to Collision Attacks on Round-Reduced Keccak. CRYPTO 2017.

Collision Attacks on Round-Reduced SHA-3

Methods	SHA3-224	SHA3-256	SHA3-384	SHA3-512	SHAKE128
Differential or SAT-based	2 (practical)	2 (practical)	-	-	*
Differential [DDS12]	4 (practical)	4 (practical)	-	-	*
Internal Diff. [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)	-
Algebriac Diff. [QSLG17]	5 (2^{101})	-	-	-	5 (practical)
Algebriac Diff. [SLG17]	5 (practical)	-	-	-	-
Differential [GLLLQS20]	5 (practical)	5 (practical)	-	-	5 (practical)

Jian Guo, Guohong Liao, Guozhen Liu, Meicheng Liu, Kexin Qiao, Ling Song: Practical Collision Attacks against Round-Reduced SHA-3. Journal of Cryptology 33(1): 228-270 (2020)

SAT-based Techniques

- Limitations of Previous Collision Attacks
  - Quick consumption of degrees of freedom in connector construction
  - Difficulty in generating colliding trails of more rounds

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- SAT-based Connector [HBDM22]

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  - Quick consumption of degrees of freedom in connector construction
  - Difficulty in generating colliding trails of more rounds
- SAT-based Connector [HBDM22]
- SAT-based Differential Trail Search [GLST22]

Collision Attacks on Round-Reduced SHA-3

Methods	Keccak-224	Keccak-256	Keccak-384	Keccak-512	SHAKE128
Differential or SAT-based	2 (practical)	2 (practical)	-	-	*
Differential [DDS12]	4 (practical)	4 (practical)	-	-	*
Internal Diff. [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)	-
Differential [GLLLQS20]	5 (practical)	5 (practical)	-	-	5 (practical)
SAT-based Diff. [HBDM22]	-	-	4 ($2^{59.64}$)	-	-

Senyang Huang, Orna Agmon Ben-Yehuda, Orr Dunkelman, Alexander Maximov: Finding Collisions against 4-Round SHA3-384 in Practical Time. TOSC 2022.

1-round SAT-based connector

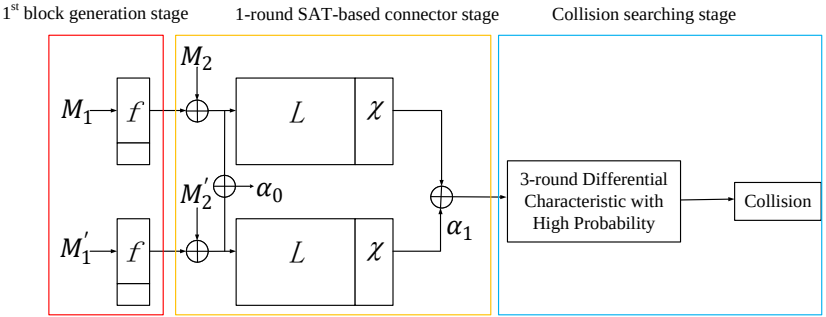


Figure: The framework of the attack

Senyang Huang, Orna Agmon Ben-Yehuda, Orr Dunkelman, Alexander Maximov: Finding Collisions against 4-Round SHA3-384 in Practical Time. TOSC 2022.

Collision Attacks on Round-Reduced SHA-3

Methods	Keccak-224	Keccak-256	Keccak-384	Keccak-512	SHAKE128
Differential or SAT-based	2 (practical)	2 (practical)	-	-	*
Differential [DDS12]	4 (practical)	4 (practical)	-	-	*
Internal Diff. [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)	-
Differential [GLLLQS20]	5 (practical)	5 (practical)	-	-	5 (practical)
SAT-based Diff. [HBDM22]	-	-	4 ($2^{59.64}$)	-	-
SAT-based Diff. and Quantum [GLST22]	$6^{\dagger}$($2^{96.15}$)	$6^{\dagger}$($2^{102.65}$)	-	-	6 ($2^{123.5}$) $6^{\dagger}$($2^{61.4}$)

† Quantum

Jian Guo, Guozhen Liu, Ling Song, Yi Tu: Exploring SAT for Cryptanalysis: (Quantum) Collision Attacks against 6-Round SHA-3. ASIACRYPT 2022.

Collision Attacks on 6-round SHA-3

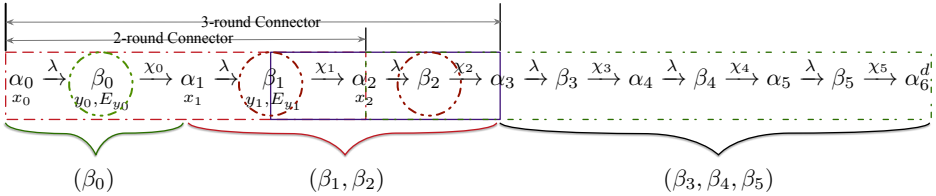


Figure: The 6-round collision attack model

Jian Guo, Guozhen Liu, Ling Song, Yi Tu: Exploring SAT for Cryptanalysis: (Quantum) Collision Attacks against 6-Round SHA-3. ASIACRYPT 2022.

Collision Attacks on Round-Reduced SHA-3

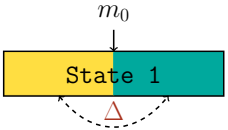
Methods	SHA3-224	SHA3-256	SHA3-384	SHA3-512	SHAKE128	SHAKE256
Differential or SAT-based	2 (practical)	2 (practical)	-	-	*	-
Differential [DDS12]	4 (practical)	4 (practical)	-	-	*	-
Internal Diff. [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)	-	-
Differential [GLLLQS20]	5 (practical)	5 (practical)	-	-	5 (practical)	-
SAT-based Diff. [HBDM22]	-	-	4 ($2^{59.64}$)	-	-	-
SAT-based Diff. and Quantum [GLST22]	$6^{\dagger}$($2^{96.15}$)	$6^{\dagger}$($2^{102.65}$)	-	-	6 ($2^{123.5}$) $6^{\dagger}$($2^{61.4}$)	-
Poly. Systems [Dinur21]	-	-	-	4^{*}($2^{263} \approx 2^{250}$ Hs)	-	-
Internal Diff. [ZHL23]	5 (2^{105})	5 (2^{105})	4 (2^{76})	4 (2^{237})	5 (2^{105})	4 (2^{76}) 5 (2^{185})

Zhongyi Zhang, Chengan Hou, Meicheng Liu: Collision Attacks on Round-Reduced SHA-3 Using Conditional Internal Differentials. EUROCRYPT 2023.

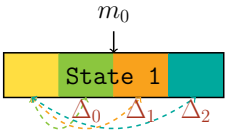
Internal Differential Cryptanalysis

- Differential cryptanalysis [BS91]:

- Internal differential cryptanalysis [Peyrin10]:



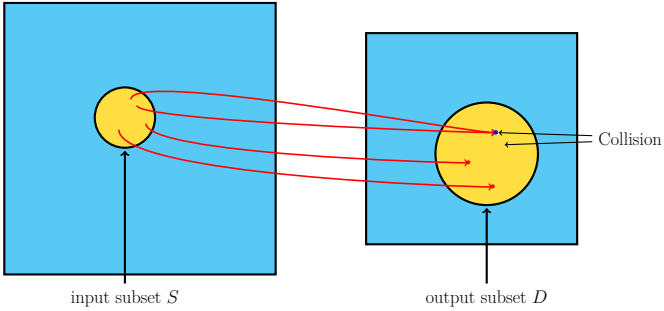
- Generalized internal differential cryptanalysis [DDS13]:



The Birthday Attack and Squeeze Attack [DDS13]

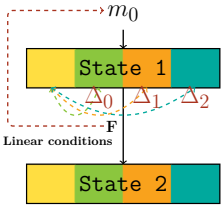
Let $H : \mathbb{F}_2^* \to \mathbb{F}_2^d$ be a hash function.

- The birthday attack
  To find a collision, $2^{d/2}$ inputs are needed.
- Squeeze attack
  H maps one input subset S into one output subset $D \subset \mathbb{F}_2^d$.
  To find a collision, $2^{n/2}$ inputs from S are needed, where $\#D = 2^n$.



Internal Differential Cryptanalysis

- Conditional internal differential cryptanalysis [ZHL23]:

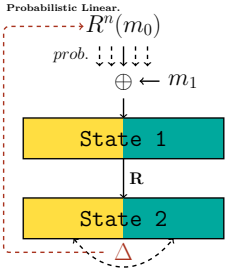


Find messages conforming 2-round internal differential characteristic by adding linear conditions to the initial state space.

Zhongyi Zhang, Chengan Hou, Meicheng Liu: Collision Attacks on Round-Reduced SHA-3 Using Conditional Internal Differentials. EUROCRYPT 2023.

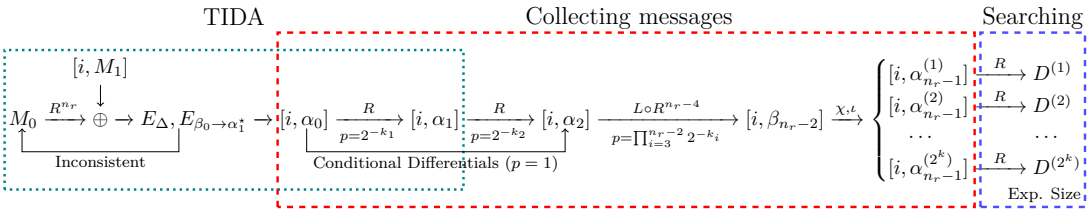
Internal Differential Attack Using Probabilistic Linearization

- Probabilistic Linearization [ZHL24]:



Find the first block message conforming target (internal) differential characteristic with a certain probability by adding LESS linear conditions to the initial internal difference space.

Zhongyi Zhang, Chengan Hou, Meicheng Liu. Probabilistic Linearization: Internal Differential Collisions in up to 6 Rounds of SHA-3. CRYPTO 2024.



5-round collision attack on SHA-3

- Select M_0 and M_1 by TIDA such that the state enters the internal difference of the second round in a given characteristic
- Calculate the internal difference of M after 4 rounds of round function and store the state into the subset $[i, v_j^{(4)}]$
- Calculate the collision subset of each subset $[i, v_j^{(4)}]$ in turn until one collision is found in a certain collision subset $D^{(j)}$

Collision Attacks on Round-Reduced SHA-3

Methods	SHA3-224	SHA3-256	SHA3-384	SHA3-512	SHAKE128	SHAKE256
Differential or SAT-based	2 (practical)	2 (practical)	-	-	-	-
Differential [DDS12]	4 (practical)	4 (practical)	-	-	-	-
Internal Diff. [DDS13]	-	5 (2^{115})	3 (practical) 4 (2^{147})	3 (practical)	-	-
Algebriac Diff. [GLLLQS20]	5 (practical)	5 (practical)	-	-	5 (practical)	-
SAT-based Diff. [HBDM22]	-	-	4 ($2^{59.64}$)	-	-	-
SAT and Quant. [GLST22]	$6^{\dagger}$($2^{96.15}$)	$6^{\dagger}$($2^{102.65}$)	-	-	6 ($2^{123.5}$) $6^{\dagger}$($2^{61.4}$)	-
Internal Diff. [ZHL23]	5 (2^{105})	5 (2^{105})	4 (2^{76})	4 (2^{237})	5 (2^{105})	4 (2^{76}) 5 (2^{185})
Probabilistic Linear. Internal Diff. [ZHL24]	5 ($2^{96.67}$)	5 ($2^{96.67}$)	5 ($2^{172.19}$)	4 ($2^{225.29}$)	5 ($2^{96.67}$)	5 ($2^{163.28}$) 6 ($2^{232.29}$)

† Quantum

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Preimage Attacks on SHA-3

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Preimage attacks and second preimage attacks

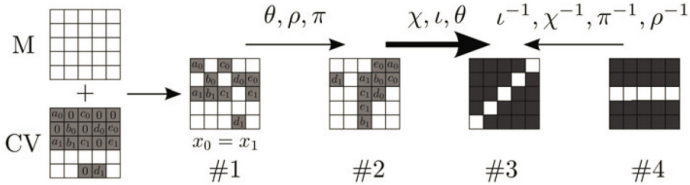
Methods	Keccak-224	Keccak-256	Keccak-384	Keccak-512
MITM [NRM11] SAT-based [HMR+12]	2 (practical)	2 (practical)	-	-
Rotational [MPS13]	4 (2^{221})	4 (2^{252})	4 (2^{378})	4 (2^{506})

María Naya-Plasencia, Andrea Röck, Willi Meier: Practical Analysis of Reduced-Round Keccak. INDOCRYPT 2011.

Pawel Morawiecki, Josef Pieprzyk, Marian Srebrny: Rotational Cryptanalysis of Round-Reduced Keccak. FSE 2013.

Preimage Attacks on 2-round Keccak

Figure: Diagram of the 2-round preimage attack. Each square represents a 64 bit lane. Each white lane is a lane known and fixed, each colored one, a not-yet-fixed lane.



María Naya-Plasencia, Andrea Röck, Willi Meier: Practical Analysis of Reduced-Round Keccak. INDOCRYPT 2011.

Preimage Attacks on Keccak

#Rounds	Variant	Time	Reference
2	Keccak-224/256	2^{33}	[Naya-PlasenciaRM11]
2	Keccak-224/256	1	Our results
2	Keccak-384/512	$2^{129}/2^{384}$	Our results
3	SHAKE128	1	Our results
3	Keccak-224/256/384	$2^{97}/2^{192}/2^{322}$	Our results
3	Keccak-512	2^{482}	Our results
3	Keccak-512	2^{506}	[MorawieckiPS13]
4	SHAKE128	2^{106}	Our results
4	Keccak-224/256	$2^{213}/2^{251}$	Our results
4	Keccak-224/256	$2^{221}/2^{252}$	[MorawieckiPS13]
4	Keccak-384/512	$2^{378}/2^{506}$	[MorawieckiPS13]

Jian Guo, Meicheng Liu, Ling Song. *Linear Structures: Applications to Cryptanalysis of Round-Reduced Keccak.* Asiacrypt 2016.

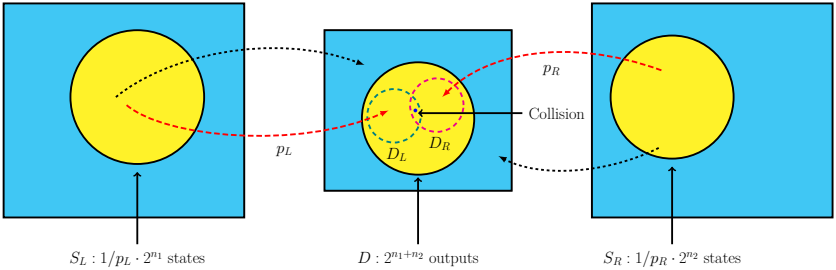
Preimage Attacks on SHA-3 up to 5 rounds

Methods	SHA3-224	SHA3-256	SHA3-384	SHA3-512	SHAKE128	SHAKE256
Rotational [MPS13]	4 (2^{221})	4 (2^{252})	4 (2^{378})	4 (2^{506})	-	-
Linear Structure [GLS16]	4 (2^{213})	4 (2^{251})	3 (2^{322})	3 (2^{482})	4 (2^{106})	4 (2^{251})
Linear Structure [LS19]	4 (2^{207})	4 (2^{239})	-	-	-	4 (2^{239})
Linear Structure [HLY21]	4 (2^{192})	4 (2^{218})	-	-	-	-
Solving Poly. [Dinur21]	4 ($2^{217\dagger}$)	4 ($2^{246\dagger}$)	4 ($2^{374\dagger}$)	4 ($2^{502\dagger}$)	-	-
MITM [QHD+23]	-	-	-	4 ($2^{504.58}$)	-	-
Squeeze-MITM [ZHL25]	4 ($2^{135.5}$) 5 ($2^{216.03\dagger}$)	4 ($2^{151.5}$) 5 ($2^{254.33\dagger}$)	4 ($2^{277.8}$)	3 ($2^{504.2}$)	4 ($2^{81.5}$) 5 ($2^{100.5}$)	4 ($2^{151.5}$) 5 ($2^{254.33\dagger}$)

$\dagger$ Bit operations

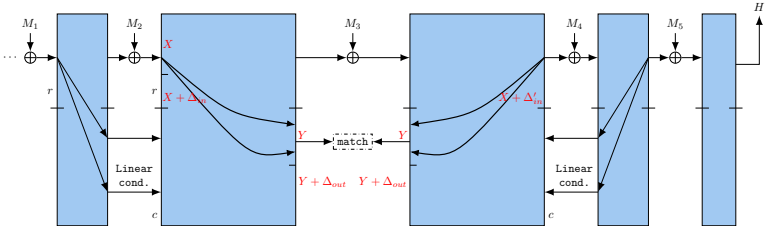
Squeeze Meet-in-the-Middle Attack [ZHL25]

- An input subset $S_L(S_R)$ is mapped with probability $p_L(p_R)$ to the output subset $D_L(D_R)$
- Use squeeze attack to find a collision in the middle subset D, $|D| = |D_L| \cdot |D_R|$
- The time complexity of the attack is $1/p_L \cdot |D_L| + 1/p_R \cdot |D_R|$

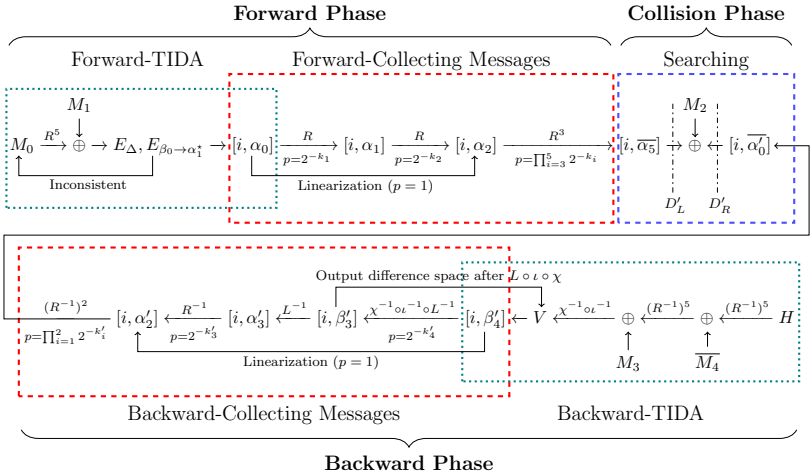


Zhongyi Zhang, Chengan Hou, Meicheng Liu. Preimage Attacks on up to 5 Rounds of SHA-3 Using Internal Differentials. EUROCRYPT 2025.

Squeeze MITM Preimage Attacks on Sponge Constructions



- $H = f(M_5 + f(M_4 + f(M_3 + f(M_2 + f(M_1)))))$
- Meet in the middle: Store $f(M_2 + f(M_1))$ in V, match $f(M_2 + f(M_1))$ and $f^{-1}(f^{-1}(f^{-1}(H) + M_5) + M_4)$ in the capacity part
- Set the value of $M_3 = f(M_2 + f(M_1)) + f^{-1}(f^{-1}(f^{-1}(H) + M_5) + M_4)$
- Preset the internal difference to launch the squeeze attack



5-round preimage attack on SHA-3

- Search for $f(f(M_0) \oplus M_1)$ and $f^{-1}(f^{-1}(f^{-1}(H) \oplus M_4) \oplus M_3)$ with the same capacity.

Preimage Attacks on SHA-3 up to 5 rounds

Methods	SHA3-224	SHA3-256	SHA3-384	SHA3-512	SHAKE128	SHAKE256
Rotational [MPS13]	4 (2^{221})	4 (2^{252})	4 (2^{378})	4 (2^{506})	-	-
Linear Structure [GLS16]	4 (2^{213})	4 (2^{251})	3 (2^{322})	3 (2^{482})	4 (2^{106})	4 (2^{251})
Linear Structure [LS19]	4 (2^{207})	4 (2^{239})	-	-	-	4 (2^{239})
Linear Structure [HLY21]	4 (2^{192})	4 (2^{218})	-	-	-	-
Solving Poly. [Dinur21]	4 ($2^{217\dagger}$)	4 ($2^{246\dagger}$)	4 ($2^{374\dagger}$)	4 ($2^{502\dagger}$)	-	-
MITM [QHD+23]	-	-	-	4 ($2^{504.58}$)	-	-
Squeeze-MITM [ZHL25]	4 ($2^{135.5}$) 5 ($2^{216.03\dagger}$)	4 ($2^{151.5}$) 5 ($2^{254.33\dagger}$)	4 ($2^{277.8}$)	3 ($2^{504.2}$)	4 ($2^{81.5}$) 5 ($2^{100.5}$)	4 ($2^{151.5}$) 5 ($2^{254.33\dagger}$)

$\dagger$ Bit operations

Outline

Introduction

Collision Aattacks on SHA-3

Preimage Attacks on SHA-3

Attacks on Keyed Mode of Keccak

Distinguishers of Keccak

Summary

Security of Keyed Modes of Keccak

- Keyed modes of Keccak
  - hash-based message authentication code (MAC)
  - stream cipher
  - authenticated encryption scheme - Ketje and Keyak

Security of Keyed Modes of Keccak

- Practical key recovery attacks

method cube attacks

- 5-round MAC
- 6-round stream cipher and Keyak
  - ⋆ the final nonlinear step χ can be inverted from the given output bits of stream cipher and Keyak

Itai Dinur, Pawel Morawiecki, Josef Pieprzyk, Marian Srebrny, Michal Straus: Cube Attacks and Cube-Attack-Like Cryptanalysis on the Round-Reduced Keccak Sponge Function. EUROCRYPT 2015: 733–761

Security of Keyed Modes of Keccak

- Practical key recovery attacks

method cube attacks

  - 5-round MAC
  - 6-round stream cipher and Keyak
    - ⋆ the final nonlinear step χ can be inverted from the given output bits of stream cipher and Keyak

- Attack more rounds with impractical complexities

method cube testers

  - key recovery attacks
    - 7-round MAC: 2^{97}
    - 7-round Keyak: 2^{76}
  - forgery attacks
    - 8-round MAC: 2^{129}
    - 7-round Keyak: 2^{65}

Itai Dinur, Pawel Morawiecki, Josef Pieprzyk, Marian Srebrny, Michal Straus: Cube Attacks and Cube-Attack-Like Cryptanalysis on the Round-Reduced Keccak Sponge Function. EUROCRYPT 2015: 733–761

Security of Keyed Modes of Keccak

- ∗ Conditional cube attack [HWX+17]
- Practical key recovery attacks
  - 6-round MAC
  - 7-round Keyak
- Key recovery attacks with impractical complexities
  - 7-round MAC: 2^{72}
  - 8-round Keyak: 2^{74}
- ⋆ Cube-attack-like cryptanalysis using MILP [SG18, BDL+19, ...]

Senyang Huang, Xiaoyun Wang, Guangwu Xu, Meiqin Wang, Jingyuan Zhao. Conditional Cube Attack on Reduced-Round Keccak Sponge Function. EUROCRYPT 2017.

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Distinguishers

- Distinguishers on Keccak hash function
  - differential
  - self-symmetry
  - algebraic
  - division property
  - no distinguishers reach 10 or more rounds

Security of Keccak

Distinguishers

- Distinguishers on Keccak hash function
  - differential
  - self-symmetry
  - algebraic
  - division property
  - no distinguishers reach 10 or more rounds
- Distinguishers on the underlying permutation of Keccak
  - zero-sum distinguishers [AM09, BC10, BCC11, DL11]
    - reach the full 24 rounds with complexity 2^{1575}
    - reach 15 rounds with complexity $< 2^{800}$
  - differential/internal differential boomerangs
    - reach 8 rounds with practical complexity [JN15]

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Preimage Attacks on SHA-3

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Summary

- Cryptanalysis of SHA-3 Hash Function
  - collision attacks: up to 6 rounds
  - preimage attacks: up to 5 rounds
  - distinguisher
    - hash function: less than 10 rounds
    - Keccak-f permutation: the full 24 rounds
  - keyed mode: up to 9 rounds
- Directions of future work
  - internal differentials in preimage and collision attacks
  - differential and algebraic techniques in collision attacks
  - partially linearization and nonlinear techniques
  - SAT-based techniques

Thank you!